

1. Quiz (15 Questions – Mix of MCQ, True/False, Short Answer)

Concept Check – Java Object & Class

Part A: Multiple Choice (5 questions)

1. What does the **new** keyword do in Java?
 - A. Declares a new class
 - B. Creates a new method
 - ☒ C. Creates a new object from a class
 - D. Deletes an object
2. Which of the following is a valid Java constructor for a class named **Dog**?
 - ☒ A. `public void Dog()`
 - B. `public Dog()`
 - C. `void Dog()`
 - D. `constructor Dog()`
3. What is the default return type of a constructor?
 - A. `int`
 - ☒ B. `void`
 - C. `String`
 - D. No return type
4. What is the output of the below code ?

```
class Constructor
{
    static int x = 1;
    public Constructor()
    {
        x = 5;
    }

    public void showX()
    {
```

```
        System.out.println("x : " + x);
    }

    public static void main(String[] args)
    {
        Constructor obj = new Constructor();
        System.out.println(x);
        obj.showX();
    }
}
```

☒ A. 5

x : 5

B. x : 5

x : 5

C. Compilation Error

D. Run time error

5. Which of these is NOT a benefit of OOP?

A. Code reuse

B. Scalability

C. Better security

☒ D. Slower performance

Part B: True/False (5 questions)

6. A Java class can have multiple constructors.

True

7. Objects in Java are created from classes.

True

8. You must always define a `main()` method inside a class to compile it.

False

9. Instance variables are declared inside methods.

False

10. The `this` keyword refers to the current object. *True*

Part C: Short Answer (5 questions)

11. Define what an object is in Java. *Object is an instance of a class that contains variables and method*
12. What is the difference between a class and an object?
A class is a blueprint used to create objects, an object is a real instance of a class
13. Why do we use constructors in a class?
use to initialize object when they are created
14. Write the code to create an object of `Car` class.
Car myCar = new Car();
15. Name two things that a class can contain.
variables & method
-



2. Code Examples



Example: Student Class

Unset

```
public class Student {  
    String name;  
    int age;  
  
    public Student(String studentName, int studentAge) {  
        name = studentName;  
        age = studentAge;  
    }  
  
    public void printDetails() {  
        System.out.println("Name: " + name + ", Age: " + age);  
    }  
  
    public static void main(String[] args) {
```

```

        Student s1 = new Student("Alya", 20);
        s1.printDetails();

        Student s2 = new Student("Zafran", 22);
        s2.printDetails();
    }
}

```

Challenge Task: Book Class (Student Practice-Fix the errors)

Unset

```

public class Book {
    String title;
    String author;
    int yearPublished;

    public Book(int yString t, String a, String tint y) {
        title = t;
        author = a;
        yearPublished = y;
    }

    public void displayBookInfo() {
        System.out.println("Title: " + title + ", Author: " +
author + ", Year: " + yearPublished);
    }

    public voidint isClassic() {
        return yearPublished < 2000;
    }

    public static void Mainmain(String[] args) {
        Book b1b1 = new Book("To Kill a Mockingbird", "Harper Lee", 1960);
displayBookInfo();
        b1.displayBookInfo();
    }
}

```

```
        b1. isClassic C)
    System.out.println("Classic? " + isClassic);
    }
}
```

Check and Fix the errors in Book.java . The output should be as below :

Output:

```
Title: To Kill a Mockingbird, Author: Harper Lee, Year: 1960
Classic? true
```

3. Student Worksheet

Title: Object Class in Java - Practice Worksheet

Section A – Fill in the Blanks

1. A class is a blueprint for creating objects.
2. An object is an instance of a class.
3. The new keyword is used to create objects.
4. A constructor has the same name as the class and no return type.
5. Use the `this` keyword to refer to the current object.

Section B – Write Code

Create a `Movie` class with:

- Fields: `title`, `genre`, `rating` (double)
- Constructor
- Method: `printMovieInfo()`

Then create 2 `Movie` objects in `main()` and call the method.

Section C – Debug the Code

Find and fix the errors:

Unset

```
public class Car {
    String model;
    int year;

    public void Car(String m, int y) {
        model = m;
        year = y;
    }

    public void printInfo() {
        System.out.println("Model: " + model + ", Year: " +
year);
    }

    public static void main(String[] args) {
        Car myCar = new Car("Honda", 2022);
myCar.printinfo();
    }    myCar.printInfo ();
}
```

Section D – Reflection (Short Answer)

1. What was easy about today's coding tasks?
fix error from the coding given.
 2. What was hard or confusing?
the way of using static with non-static function.
 3. Rate your confidence (1–10) with creating Java objects.
8.
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